STUDYLAB HUB · AP & COLLEGE BIOLOGY · UNIT 1 (CHAPTERS 1–5)

Unit 1: Chemistry of Life

Scientific Method · Water & Hydrogen Bonding · Carbon · Carbohydrates, Lipids, Proteins & Nucleic Acids

AP & College Biology — Unit 1 (Chapters 1–5): Chemistry of Life

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1. Unit 1 Roadmap (Chapters 1–5)

Chapter / CED Topic	Core Concept	What Exams Actually Test
Ch. 1 · The Study of Life	Scientific method, hypotheses, null vs. alternative hypotheses, experimental controls	Identifying independent/dependent variables, positive vs. negative controls, and interpreting error bars (±2 SEM)
Ch. 2 · Water & Life (CED 1.1)	Polar covalent O–H bonds, intermolecular hydrogen bonding, emergent properties of water	Predicting what happens when polarity or H-bonding is disrupted (cohesion, adhesion, surface tension, high specific heat, evaporative cooling, ice density, pH & buffers)
Ch. 3–4 · Carbon & Functional Groups (CED 1.2)	Tetravalent carbon (4 valence electrons), functional groups, dehydration synthesis vs. hydrolysis	Recognizing hydroxyl, carbonyl, carboxyl (acidic), amino (basic), sulfhydryl (disulfide bridges), phosphate (negative charge/ATP), and methyl (nonpolar/epigenetic) groups
Ch. 5 · Biological Macromolecules (CED 1.3–1.6)	Carbohydrates, Lipids, Proteins (1°–4° structure), Nucleic Acids (DNA vs. RNA)	Directionality (5′ → 3′ vs. N-terminus → C-terminus), monomer-to-polymer bond types, and how R-group mutations alter protein folding

2. Chapter 1: Experimental Design & Statistical Reasoning

1. **Null Hypothesis (H_0) vs. Alternative Hypothesis (H_a)**:

- **Null Hypothesis (H_0)**: States there is **no statistically significant difference** between treatment groups (any observed difference is due to random sampling chance).
- **Alternative Hypothesis (H_a)**: States that the independent variable causes a real, measurable effect on the dependent variable.

2. **Controls on Free-Response Questions (FRQs)**:

- **Negative Control**: Group receiving no treatment (or solvent/placebo only) to establish the baseline and rule out confounding variables.
- **Positive Control**: Group receiving a treatment known to produce the expected effect, confirming that the experimental setup and reagents work.

3. **Standard Error of the Mean (SEM)**:

- Error bars showing **±2 SEM** represent an approximate **95% confidence interval**.
- When the **±2 SEM** error bars between two groups **do not overlap**, the difference between their means is statistically significant.

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3. Chapter 2 (CED 1.1): Structure of Water & Hydrogen Bonding

- **Inside one water molecule**: Oxygen is much more electronegative than hydrogen (O ≈ 3.44 vs. H ≈ 2.20). Shared electrons spend more time near oxygen, creating a **polar covalent bond** with a partial negative charge (δ^-) on oxygen and partial positive charges (δ^+) on each hydrogen.
- **Between water molecules**: The δ^+ hydrogen of one water molecule attracts the δ^- oxygen lone pairs of a neighboring water molecule, forming an intermolecular **hydrogen bond**.

Property of Water	Molecular Mechanism	Biological Consequence
Cohesion	Hydrogen bonds hold water molecules to **each other**	Creates high surface tension; pulls continuous water column upward during transpiration in plant xylem
Adhesion	Hydrogen bonds form between water and **other polar/charged surfaces** (e.g., cellulose cell walls)	Counteracts gravity in narrow plant xylem vessels (capillary action)
High Specific Heat	Absorbed heat must first break many hydrogen bonds before raising average kinetic energy (temperature)	Buffers ocean temperatures and keeps intracellular temperature stable
High Heat of Vaporization	Breaking hydrogen bonds for liquid → gas requires substantial thermal energy	Evaporative cooling (sweating in mammals, transpiration in leaves) removes excess body/leaf heat
Ice Floats (Solid Less Dense)	At 0°C, water locks into an open crystalline lattice where each molecule hydrogen-bonds to 4 neighbors	Ice forms an insulating surface layer on lakes and ponds so aquatic life survives underneath in winter
Versatile Polar Solvent	Polar water molecules form hydration shells around ions (Na^+, Cl^-) and polar solutes	Dissolves blood plasma solutes and cytosolic reactants (hydrophilic vs. hydrophobic interactions)

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4. Chapters 3–5 (CED 1.2–1.6): Dehydration Synthesis, Hydrolysis & The 4 Macromolecules

- **Dehydration Synthesis (Condensation)**: Joins monomers into a polymer by **removing one H_2O molecule** per covalent bond formed (endergonic; requires enzymes and energy).
- **Hydrolysis**: Breaks a polymer back into monomers by **adding one H_2O molecule** across the covalent bond (exergonic; digestive breakdown).

Macromolecule	Elements	Monomer / Subunit	Covalent Linkage	Key Examples & Directionality
Carbohydrates	C, H, O (1 : 2 : 1 ratio)	Monosaccharides (glucose, fructose, ribose)	**Glycosidic linkage** (α-1,4 in starch/glycogen vs. β-1,4 in cellulose)	Starch & glycogen (energy storage); cellulose & chitin (structural support; humans lack the enzyme cellulase to hydrolyze β-1,4 linkages)
Lipids *(not true polymers)*	C, H, O (high H:O ratio; phospholipids also contain P & N)	Glycerol + fatty acids (or 4 fused carbon rings in steroids)	**Ester linkage**	Saturated (no C=C double bonds, straight chains, solid at room temp) vs. Unsaturated (*cis* C=C double bonds create kinks, liquid/fluid membranes); **Phospholipids** are amphipathic (hydrophilic head + hydrophobic tails)
Proteins	C, H, O, N, S	Amino acids (central α-carbon, amino group, carboxyl group, R-group)	**Peptide bond** (carboxyl C to amino N)	Synthesized **N-terminus ($-NH_3^+$) → C-terminus ($-COO^-$)**; R-groups are nonpolar/hydrophobic, polar/hydrophilic, or ionic/charged (acidic/basic)
Nucleic Acids	C, H, O, N, P	Nucleotides (5-carbon pentose sugar + nitrogenous base + phosphate)	**Phosphodiester linkage** (3′-OH of one sugar to 5′-phosphate of next)	Synthesized **5′ → 3′**; antiparallel DNA strands held by hydrogen bonds (**A=T** has 2 H-bonds; **G≡C** has 3 H-bonds, so higher GC% = higher melting temperature)

5. Protein Folding Hierarchy (1° → 4°) & Mutation Analysis

1. **Primary (1°) Structure**: Exact linear sequence of amino acids joined by covalent **peptide bonds** (determined directly by mRNA codons).
2. **Secondary (2°) Structure**: Local α-helices and β-pleated sheets stabilized by **hydrogen bonds along the polypeptide backbone** (between carbonyl O and amide H — *not* R-groups).
3. **Tertiary (3°) Structure**: Overall 3D fold of a single polypeptide driven by **R-group (side-chain) interactions**: hydrophobic clustering inside the interior, hydrogen bonds, ionic salt bridges, van der Waals attractions, and covalent **disulfide bridges (–S–S–)** between cysteine residues.
4. **Quaternary (4°) Structure**: Assembly of **two or more polypeptide chains** into one functional protein complex (e.g., hemoglobin's 2α + 2β subunits).

> **Classic AP FRQ Mutation Rule**:
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> - Replacing a hydrophobic amino acid (e.g., Valine) with another similarly sized hydrophobic amino acid (e.g., Leucine) usually preserves 3° folding (conservative substitution).

> - Replacing a charged/polar amino acid (e.g., negatively charged Glutamate) with a nonpolar hydrophobic amino acid (e.g., Valine, as in sickle-cell hemoglobin) changes local polarity, alters 3° and 4° folding, and disrupts protein function.

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6. Worked Exam Practice (MCQs + Scored FRQ)

Practice Multiple-Choice

1. **A double-stranded DNA molecule from Organism A contains 18% Adenine (A). What percentage of the bases in this DNA molecule are Cytosine (C), and how does its thermal stability compare to a DNA molecule with 35% Adenine?**

- **Answer**: By Chargaff's rules, %A = %T = 18% (36% total), leaving 64% for G + C, so **%C = 32%**. Because G≡C base pairs form **3 hydrogen bonds** (vs. 2 for A=T), Organism A's DNA (64% GC) has **higher thermal stability** (denatures at a higher temperature) than the 35% A DNA (30% GC).

2. **A transmembrane protein spans the hydrophobic core of the phospholipid bilayer. Which amino acid substitution in its membrane-spanning segment is most likely to disrupt its membrane insertion?**

- **Answer**: Substituting a **nonpolar/hydrophobic** amino acid (e.g., Phenylalanine or Isoleucine) with a **charged/hydrophilic** amino acid (e.g., Lysine or Aspartate), which is energetically unfavorable inside the nonpolar fatty-acid tail interior.

Worked Free-Response Question (FRQ)

Prompt: Researchers measure the catalytic rate of a bacterial protease from 10°C to 80°C. Activity peaks at 37°C and drops to zero at 70°C.

- **(a) Identify the specific covalent bond broken by the protease when it digests a polypeptide, and state whether water is consumed or produced.**
- **(b) Explain at the molecular level why heating the enzyme to 70°C reduces activity to zero even though its primary (1°) amino acid sequence remains intact.**

Scoring Rubric:

- **(a)** Breaks covalent **peptide bonds** between amino acids; **water (H_2O) is consumed** (added across the bond during hydrolysis).
- **(b)** High thermal energy disrupts weak noncovalent interactions (**hydrogen bonds, ionic salt bridges, and hydrophobic interactions**) among R-groups and backbone atoms, unfolding the **2° and 3° structure (denaturation)** and distorting the active site without cleaving covalent peptide bonds (1° structure).